

JORGE

ANAIS VILCHEZ

CONTACT

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EDUCATION

Doctorate in Astrophysics and Astroinformatics <i>Universidad de Antofagasta</i> Antofagasta, Chile PhD candidate. Thesis: "Characterization of Sagittarius dwarf galaxy at low Galactic latitudes"	2022 - (2025)
Master in Astronomy <i>Universidad de Antofagasta</i> Antofagasta, Chile Thesis: "Search and characterization of star clusters in the far end of the Galactic bar".	2018 - 2020
Licentiate in Astronomy <i>Pontificia Universidad Católica de Chile</i> Santiago, Chile Graduation activity: "A new astrometric software for the Swope Telescope", developed at Las Campanas Observatory.	2008 - 2014

EXPERIENCE

Early Career Visitor Programme <i>Characterization of new Sagittarius Dwarf Galaxy Clusters</i> ESO, Chile Participated in a project led by Dr. Elisa R. Garro combining deep HST photometry (WFC3, F555W and F814W) and FLAMES+UVES spectroscopy (98 hrs, PI: Garro) to precisely determine ages and chemical compositions of Sagittarius dwarf galaxy globular clusters. Goals include resolving the age-metallicity relation, testing formation scenarios during Sgr-Milky Way interaction, and comparing with theoretical models to reconstruct Sgr's evolutionary history.	04/2025 - Present
Research Project <i>Variable stars in the Sagittarius dwarf spheroidal galaxy</i> NOIRLab Collaboration with NOIRLab researcher Kathy Vivas to detect and classify variable stars (focus on δ Scuti and RR Lyrae stars) in the central regions of Sagittarius galaxy using multiepoch DECam data.	09/2024 - Present
Data Scientist <i>Grupo Kaufmann AWTO</i> Chile and Brasil Design and implement machine learning models, perform exploratory data analysis, communicate insights and recommendations to stakeholders.	08/2021 - 04/2022
Master's Thesis <i>Search and characterization of star clusters in the far end of the Galactic bar</i> CITEVA, Universidad de Antofagasta Development of a methodology based on unsupervised machine learning to search for young cluster candidates at the far end of the Galactic bar. This work allowed us to identify three new cluster candidates at a highly extincted and crowded region of the Milky Way. Under the supervision of Dr. Sebastián Ramírez and Dra. Karla Peña Ramírez.	03/2020 - 11/2020

Research Assistant	03/2020 - 11/2020
<i>Spectroscopy models for planetary atmospheres</i> CITEVA, UA	
Study on transmission spectroscopy <i>in silico</i> models for Earth-like and Mini-Neptunes planetary atmospheres. The result of this research was used for a JWST Time Proposal (#2684, cycle 1). Under the supervision of Dr. Jeremy Tregloan-Reed.	
Observer	2015 - 2020
<i>1M2H Project, UC Santa Cruz</i> Las Campanas Observatory, Atacama, Chile	
Photometric transients follow-up using Swope Telescope. Responsible for data acquisition, data quality assurance, and system troubleshooting. Under the supervision of Dr. Ryan Foley.	
Research Assistant	01/2016 - 06/2016
<i>Data reduction</i> Instituto de Astronomía, Universidad Católica del Norte, Antofagasta, Chile	
Data reduction of VLT FORS and Magellan MagE data. Under the supervision of Dr. Christian Moni Bidin.	
Observer	2014 - 2015
<i>Carnegie Supernova Project II</i> Las Campanas Observatory, Atacama, Chile	
Observations of low-redshift supernovae using the Swope Telescope at Las Campanas Observatory. Under the supervision of Dr. Mark M. Phillips.	
Research Assistant	2014 - 2015
<i>Sonification of medical images</i> Instituto de Música, Pontificia Universidad Católica de Chile, Santiago, Chile	
Development of a sonification tool to assist radiologists in the detection of breast cancer. In collaboration with Dr. Rodrigo Cádiz and Dr. Patricio de la Cuadra.	

SKILLS

Software	Proficient in developing data analysis, reduction, and visualization using Python and its main associated modules (Astropy, Matplotlib, Numpy, Pandas, Plotly, PyTorch, Seaborn, SciPy). Astronomical software: IRAF, DS9, TOPCAT. Other technologies: Bash, Docker, Git, $\LaTeX$, PostgreSQL, R. Github repo: https://github.com/jorgeanais
Languages	- Spanish (native) - English (professional competence)

TEACHING EXPERIENCE

Instructor	2022 - Present
<i>Duoc UC, School of Informatics and Telecommunications</i> Santiago, Chile	
Teaching Machine Learning and Deep Learning courses for computer engineering students.	
Teaching Assistant	2010 - 2014
<i>Pontificia Universidad Católica de Chile</i> Santiago, Chile	
Courses: Introduction to Physics for Biological Sciences, Static and Dynamics for Civil Engineers, Thermodynamics, Optics, Astronomical Instrumentation and Electricity and Magnetism.	

OBSERVATIONAL EXPERIENCE

Swope 1m Telescope	2014 - 2020
<i>Las Campanas Observatory</i> Atacama, Chile	
More than 300 nights observed.	

	Chakana 0.6m Telescope <i>UA Ckoirama Observatory</i> Antofagasta, Chile 10 nights observed in total.	2018 - 2020
	Zeiss 1.23m Telescope <i>Calar Alto Astronomical Observatory</i> Spain (remote) 8 nights observed in total.	2020
	PUC 50cm Telescope / PUCHEROS Spectrograph <i>PUC Santa Martina Observatory</i> Santiago, Chile 10 nights observed in total.	2013 - 2014
FURTHER STUDIES		
	Visual Recognition with Deep Learning (CC66T) <i>Universidad de Chile</i> Santiago, Chile	2024
	Natural Language Processing (CC66Q) <i>Universidad de Chile</i> Santiago, Chile	2024
	Diplomature in Data Science <i>Universidad del Desarrollo</i> Santiago, Chile	2021
	Diplomature in Software Development <i>Duoc UC</i> Santiago, Chile	2018
	Diplomature in Astroengineering <i>Universidad de Antofagasta</i> Antofagasta, Chile	2016
AWARDS	• 2024-2026 Full Scholarship for advanced human capital in Chile, Agencia Nacional de Investigación y Desarrollo de Chile. • 2022-2023 Academic Excellence Scholarship for PhD students, Universidad de Antofagasta. • 2019 Full scholarship to attend La Serena School for Data Science, AURA Observatory, Chile.	
CONFERENCES & SCHOOLS		
	12th VVV Science Meeting: the VVX synergy with next-generation Milky Way studies <i>Conference</i> Astrophysics Group at the Universidade Federal de Santa Catarina in Florianópolis, Brazil Oral: <i>Unveiling the Sagittarius Dwarf Galaxy: Mapping Its Core and Streams at Low Galactic Latitudes</i>	04/2025
	Summer School in Statistics for Astronomers <i>Center for Astrostatistics at The Pennsylvania State University</i>	07/2024
	First LSST Latin American Meeting (LSST@LATAM): Catalyzing Research Collaborations <i>Conference</i> La Serena, Chile	03/2024
	XVII Latin American Regional IAU Meeting <i>Conference</i> Montevideo, Uruguay Poster contribution: <i>Unraveling the Sagittarius Dwarf Spheroidal Galaxy at Low Galactic latitudes</i>	03/2023

First Latinamerican Virtual Observatory School <i>Spanish Virtual Observatory Online</i>	03/2023
Pycon Chile 2021 <i>Conference Online (https://pycon.cl/2021/)</i> Contributed talk: <i>Search for star clusters using unsupervised machine learning</i>	2021
LARIM 2019: XVI Latin American Regional IAU Meeting <i>Conference Antofagasta, Chile</i>	2019
Star Form Mapper 2019 <i>Conference York, United Kingdom</i> Contributed poster: <i>Massive Open clusters in VVV data using unsupervised clustering algorithms</i>	2019
La Serena School for Data Science: Applied Tools for Data-driven Sciences <i>AURA La Serena, Chile</i>	2019

PUBLICATIONS

As first author

- Anais, J., Alonso-García, J., et. al *Unraveling the Sagittarius dwarf spheroidal galaxy at low Galactic latitudes*. Estimated submission 2025.

As coauthor

- Uddin S. A., Burns C. R., Phillips M. M., Suntzeff N. B., Freedman W. L., Brown P. J., Morrell N., et al., including **Anais, J.**, *Carnegie Supernova Project I and II: Measurements of H_0 Using Cepheid, Tip of the Red Giant Branch, and Surface Brightness Fluctuation Distance Calibration to Type Ia Supernovae*, 2024, ApJ, 970, 72. doi:10.3847/1538-4357/ad3e63
- Morrell N., Phillips M. M., Folatelli G., Stritzinger M. D., Hamuy M., et al., including **Anais, J.**, *Optical Spectroscopy of Type Ia Supernovae by the Carnegie Supernova Projects I and II*, 2024, ApJ, 967, 20. doi:10.3847/1538-4357/ad38af
- Ashall C., Lu J., Hsiao E. Y., Hoeflich P., Phillips M. M., Galbany L., Burns C. R., et al. including **Anais, J.**, *Carnegie Supernova Project: The First Homogeneous Sample of Super-Chandrasekhar-mass/2003fg-like Type Ia Supernovae*, 2021, ApJ, 922, 205. doi:10.3847/1538-4357/ac19ac,
- Armstrong P., Tucker B. E., Rest A., et al. including **Anais, J.**, *SN2017jgh: a high-cadence complete shock cooling light curve of a SN IIb with the Kepler telescope*, 2021, MNRAS, 507, 3125. doi:10.1093/mnras/stab2138
- Lu J., Ashall C., Hsiao E. Y., et al. including **Anais, J.**, *ASASSN-15hy: An Underluminous, Red 03fg-like Type Ia Supernova*, 2021, ApJ, 920, 107. doi:10.3847/1538-4357/ac1606,
- Tregloan-Reed J., Otarola A., Unda-Sanzana E., Haeussler B., Gaete F., Colque J. P., González-Fernández C., et al. including **Anais, J.**, *Optical-to-NIR magnitude measurements of the Starlink LEO Darksat satellite and effectiveness of the darkening treatment*, 2021, A&A, 647, A54. doi:10.1051/0004-6361/202039364
- Stritzinger M. D., Taddia F., Fraser M., et al. including **Anais, J.**, *The Carnegie Supernova Project II. Observations of the intermediate-luminosity red transient SNhunt120*, 2020, A&A, 639, A103. doi:10.1051/0004-6361/202038018
- Tregloan-Reed J., Otarola A., Ortiz E., Molina V., **Anais J.**, González R., Colque J. P., et al., *First observations and magnitude measurement of Starlink's Darksat*, 2020, A&A, 637, L1. doi:10.1051/0004-6361/202037958
- Phillips M. M., Contreras C., Hsiao E. Y., Morrell N., Burns C. R., Stritzinger M., Ashall C., et al. including **Anais, J.**, *Carnegie Supernova Project-II: Extending the Near-infrared Hubble Diagram for Type Ia Supernovae to $z \sim 0.1$* , 2019, PASP, 131, 014001. doi:10.1088/1538-3873/aae8bd
- Burns C. R., Parent E., Phillips M. M., Stritzinger M., Krisciunas K., Suntzeff N. B., Hsiao E. Y., et al. including **Anais, J.**, *The Carnegie Supernova Project: Absolute Calibration and the Hubble Constant*, 2018, ApJ, 869, 56. doi:10.3847/1538-4357/aae51c
- Krisciunas K., Contreras C., Burns C. R., Phillips M. M., Stritzinger M. D., Morrell N., Hamuy M., et al. including **Anais, J.**, *VizieR Online Data Catalog: The CSP (DR3): photometry of low- z SNe Ia (Krisciunas+, 2017)*, 2018, yCat, J/AJ/154/211
- Stritzinger M. D., Anderson J. P., Contreras C., Heinrich-Josties E., Morrell N., Phillips M. M., **Anais J.**, et al., *The Carnegie Supernova Project I. Photometry data release of low-redshift stripped-envelope supernovae, The Carnegie Supernova Project. I. Third Photometry Data Release of Low-redshift Type Ia Supernovae and Other White Dwarf Explosions*, 2018, A&A, 609, A134. doi:10.1051/0004-6361/201730842
- Coulter D. A., Kilpatrick C. D., Foley R. J., **Anais J.**, Phillips M. M., Drout M. R., Madore B. F., et al., *Swope Photometric Observations of SN 2017cbv = DLT17u*, 2017, ATel, 10167